the point of pivot as shown in Figure 5.15. If the rod has to remain stationary in horizontal position, it should be in translational and rotational equilibrium. Then, both the net force and net torque must be zero.

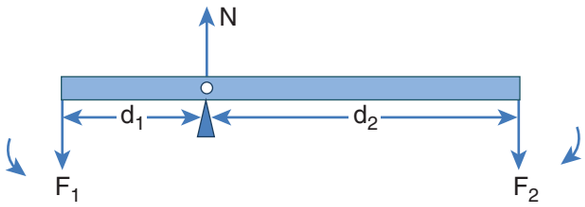


Figure 5.15 Principle of Moments

For translational equilibrium,
net force has to be zero, $-F_1 + N - F_2 = 0$

$$N = F_1 + F_2$$

For rotational equilibrium,
net torque has to be zero, $d_1F_1 - d_2F_2 = 0$

$$d_1F_1 = d_2F_2 \qquad (5.33)$$

The above equation represents the *principle of moments*. This forms the principle for beam balance used for weighing goods with the condition $d_1 = d_2$; $F_1 = F_2$. We can rewrite the equation 5.33 as,

$$\frac{F_1}{F_2} = \frac{d_2}{d_1} \qquad (5.34)$$

If F_1 is the load and F_2 is our effort, we get advantage when, $d_1 < d_2$. This implies that $F_1 > F_2$. Hence, we could lift a large load with small effort. The ratio $\left(\frac{d_2}{d_1}\right)$ is called mechanical advantage of the simple lever. The pivoted point is called fulcrum.

$$\text{Mechanical Advantage}(MA) = \frac{d_2}{d_1} \qquad (5.35)$$

There are many simple machines that work on the above mentioned principle.

5.3.4 Centre of Gravity

Each rigid body is made up of several point masses. Such point masses experience gravitational force towards the centre of Earth. As the size of Earth is very large compared to any practical rigid body we come across in daily life, these forces appear to be acting parallelly downwards as shown in Figure 5.16.

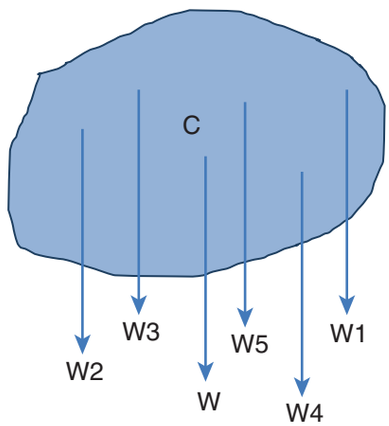


Figure 5.16. Centre of gravity

The resultant of these parallel forces always acts through a point. This point is called centre of gravity of the body (with respect to Earth). The *centre of gravity of a body is the point at which the entire weight of the body acts irrespective of the position and orientation of the body*. The centre of gravity and centre of mass of a rigid body coincide when the gravitational field is uniform across the body. The concept of gravitational field is dealt in Unit 6.

UNIT-5(XI-Physics_Vol-1).indd 231 06-01-2022 17:46:58